

The sixth cumulant of the even-monomial Gibbs weight

Tide retrospective, laplace seabed

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1 Setting

The laplace monomial track carries a cumulant ladder against the symmetric weight $\exp(-t x^{2k}/(2k)!)$: the variance κ_2 and the excess kurtosis κ_4 , each in Gamma closed form and with its $t^{-\dots}$ asymptotic scaling. This tide adds the next even rung, the sixth cumulant κ_6 , completing the ladder at the clean frontier (the general even cumulant would require set-partition / Bell-polynomial machinery).

2 The thing formalised

For a symmetric weight the mean and all odd moments vanish, so $\kappa_6 = \langle x^6 \rangle - 15\langle x^4 \rangle \langle x^2 \rangle + 30\langle x^2 \rangle^3$, and

$$\kappa_6 = \left(\frac{(2k)!}{t}\right)^{3/k} (r_3 - 15 r_2 r_1 + 30 r_1^3), \quad r_j := \frac{\Gamma((2j+1)/(2k))}{\Gamma(1/(2k))},$$

hence $\kappa_6 \sim K_6(k) t^{-3/k}$ at atTop. The Lean file gives the closed form `monomial_sixth_cumulant`, the constant `sixthCumulantConst`, and the three asymptotic rungs. At $k = 1$ the bracket is $15 - 15 \cdot 3 + 30 = 0$: a Gaussian has vanishing sixth cumulant.

3 Proof strategy

The closed form is the three-moment mirror of the kurtosis proof. Rewrite x^6, x^4, x^2 into the even form $x^{2 \cdot 3}, x^{2 \cdot 2}, x^{2 \cdot 1}$ (definitional), substitute the even-moment closed form at $j = 3, 2, 1$, and reduce everything to the common base $B := ((2k)!/t)^{1/k}$. Two power folds do this: $((2k)!/t)^{2/k} = B^2$ (as in the kurtosis) and $((2k)!/t)^{3/k} = B^3$. The cube is the one new step: `Real.rpow_natCast` turns the natural power B^3 into the real power $B^{(3:\mathbb{R})}$, and `Real.rpow_mul` collapses $(base^{1/k})^3 = base^{(1/k) \cdot 3}$; a `push_cast` and `ring` match the exponent. With both folds the goal is polynomial in B and the Gamma ratios (opaque atoms), so `ring` closes it. The three asymptotic rungs are the constant-sequence and reflexivity arguments reused verbatim from the moment and kurtosis asymptotics.

4 Roadblocks and resolutions

None; the tide compiled on the first attempt. The only forethought was including an `Authors:` line in the header from the outset, having learned the previous tide that the `style.header` linter flags its absence.

5 What was Mathlib, what was new

Mathlib supplied the `Real.rpow` algebra (including `rpow_natCast`/`rpow_mul` for the cube) and the `IsEquivalent/Tendsto` API. The closed-form even moments were already in the seabed. New here is the sixth cumulant, its constant, and its scaling — the last clean rung of the monomial cumulant ladder. No GPT consult was needed.